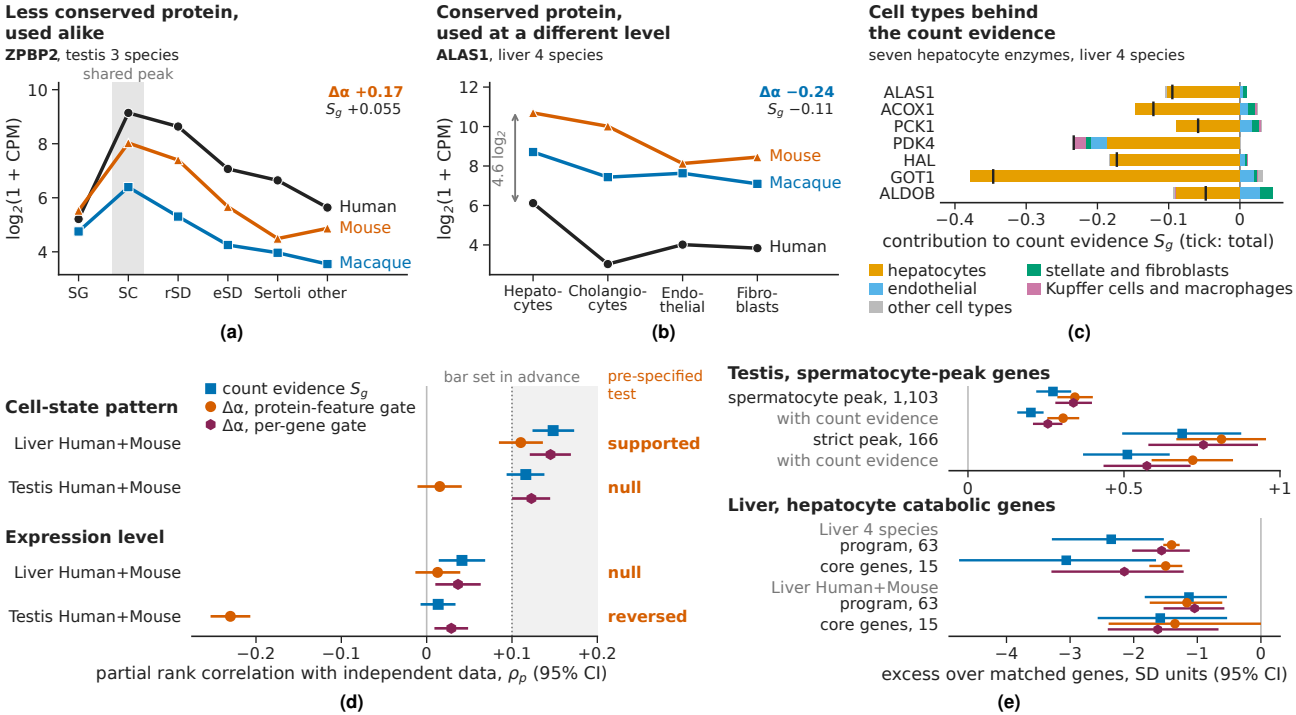

Less conserved protein, used alike
ZPBP2, testis 3 species
shared peak
Δα +0.17
$S_g$ +0.055
log2(1 + CPM)
Human
Mouse
Macaque
SG
SC
rSD
eSD
Sertoli
other
(a)
Conserved protein, used at a different level
ALAS1, liver 4 species
Δα −0.24
$S_g$ −0.11
4.6 log2
Mouse
Macaque
Human
Hepato-cytes
Cholangio-cytes
Endo-thelial
Fibro-blasts
(b)
Cell types behind the count evidence
seven hepatocyte enzymes, liver 4 species
ALAS1
ACOX1
PCK1
PDK4
HAL
GOT1
ALDOB
contribution to count evidence $S_g$ (tick: total)
hepatocytes
endothelial
other cell types
stellate and fibroblasts
Kupffer cells and macrophages
(c)
count evidence $S_g$
Δα, protein-feature gate
Δα, per-gene gate
bar set in advance
pre-specified test
Cell-state pattern
Liver Human+Mouse
supported
Testis Human+Mouse
null
Expression level
Liver Human+Mouse
null
Testis Human+Mouse
reversed
partial rank correlation with independent data, $\rho_p$ (95% CI)
(d)
Testis, spermatocyte-peak genes
spermatocyte peak, 1,103
with count evidence
strict peak, 166
with count evidence
Liver, hepatocyte catabolic genes
Liver 4 species
program, 63
core genes, 15
Liver Human+Mouse
program, 63
core genes, 15
excess over matched genes, SD units (95% CI)
(e)